MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL SEMESTER 2025
CHAPTER 4: DIFFERENTIATION

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Goal of this section:

(1) Definition of derivatives
(2) Basic Properties and Chain Rule
(3) Relative extremum via derivatives
(4) The Mean Value Theorem and applications
(5) Taylor's Theorem and applications

Reference books:

- Robert G. Bartle, Donald R. Sherbert: Introduction to Real Analysis (Wiley, 2011, Fourth Edition)
- Kenneth A. Ross: Elementary Analysis: The Theory of Calculus (Springer, 2013, Second Edition)

1. The Derivative

1.1. Definition of derivatives.

Definition 1.1. (1) Let $I \subset \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. A real number L is said to the derivative of f at c if for any given $\epsilon > 0$, there exists $\delta > 0$ such that for any $x \in I$ satisfying $0 < |x - c| < \delta$,

$$\left| \frac{f(x) - f(c)}{x - c} - L \right| < \epsilon.$$

We say that f is differentiable at $x = c$, and we write $f'(c) = L$.

(2) Equivalently, the derivative of f at c is equal to L if and only if $\lim_{x\to c} \frac{f(x)-f(c)}{x-c}$ exists, and

$$L = \lim_{x\to c} \frac{f(x) - f(c)}{x - c}.$$

Exercise 1.2. Find the derivative of the following functions at 0 from the limit definitions:

(1) $f(x) = x$

(2) $f(x) = x^3$

1.2. **Continuity of a differentiable function.**

Theorem 1.3. *(differentiable ⇒ continuous) Let $f : I \to \mathbb{R}$ be differentiable at c. Then f is continuous at c.*

Proof. For any $x \in I$ with $x \neq c$,

$$f(x) = \frac{f(x) - f(c)}{x - c} \cdot (x - c) + f(c).$$

Since $\lim_{x\to c} \frac{f(x)-f(c)}{x-c}$ and $\lim_{x\to c}(x - c)$ and $\lim_{x\to c} f(c)$ exist, $\lim_{x\to c}\ f(x)$ exist. Then,

$$\lim_{x\to c}\ f(x) = \lim_{x\to c}\ \left(\frac{f(x) - f(c)}{x - c} \cdot (x - c) + f(c)\right) = f'(c) \cdot 0 + f(c) = f(c).$$

Hence, f is continuous at c. □

Example 1.4. Let $f : \mathbb{R} \to \mathbb{R}$ given by:

$$f(x) = \begin{cases} 1 & \text{if } x \geq 1 \\ 0 & \text{if } x < 1 \end{cases}$$

Then f is not continuous at $x = 0$ and so is not differentiable at $x = 0$.

Example 1.5. The converse of Theorem 1.3 is not true. Let $f(x) = |x|$. Then

$$\frac{f(x) - f(0)}{x - 0} = \frac{|x|}{x} = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

Thus $\lim \frac{f(x)-f(0)}{x-0}$ does not exist, and so f is not differentiable at 0. We have seen in Chapter 3 that f is continuous on $\mathbb{R}$, in particular at 0.

1.3. **Useful derivatives.** We shall use the following derivatives without showing the differentiability in details:

(1) Let $f(x) = b$ be a constant function. Then $f'(x) = 0$.
(2) Let $f(x) = x^n$ for $n \in \mathbb{Z}$ and $n \neq 0$. Then $f'(x) = nx^{n-1}$. (Remark: The integral power can also be replaced by some rational powers, see Exercise 2.7.)
(3) Let $f(x) = \sin(x)$. Then $f'(x) = \cos(x)$.
(4) Let $f(x) = \cos(x)$. Then $f'(x) = -\sin(x)$.
(5) Let $f(x) = e^x$. Then $f'(x) = e^x$.
(6) Let $f : (0, \infty)$ given by $f(x) = \ln(x)$. Then $f'(x) = \frac{1}{x}$.

1.4. **More examples.**

Exercise 1.6. Determine if the following functions

(1) $f(x) = x|x|$;
(2) $f(x) = e^{|x|}$

are differentiable at $x = 0$.

Exercise 1.7. Define

$$f(x) = \begin{cases} x^2 & \text{if } x \text{ is rational} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$$

Determine the point of differentiability. Find $f'(c)$ if f is differentiable at c.

2. Basic Properties of Differentiable Functions

2.1. **Basic properties.**

Theorem 2.1. *Let $I \subset \mathbb{R}$ be an interval. Let $c \in I$. Let $f, g : I \to \mathbb{R}$ be differentiable functions at c. Then the following properties hold:*

(a) *For $\alpha \in \mathbb{R}$, αf is differentiable at c, and $(\alpha f)'(c) = \alpha(f'(c))$.*
(b) *$f + g$ is differentiable at c, and $(f + g)'(c) = (f' + g')(c)$.*
(c) *fg is differentiable at c, and $(fg)'(c) = f(c)g'(c) + f'(c)g(c)$.*
(d) *If $g(c) \neq 0$, then $\frac{f}{g}$ is differentiable at c and $\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}$.*

Proof. (a) and (b) are simple exercises from definitions.

We now consider (c).

$$\begin{aligned} \lim_{x\to c} \frac{f(x)g(x) - f(c)g(c)}{x - c} &= \lim_{x\to c} \frac{f(x)g(x) - f(c)g(x) + f(c)g(x) - f(c)g(c)}{x - c} \\ &= \lim_{x\to c} \left(g(x)\frac{f(x) - f(c)}{x - c} + \frac{g(x) - g(c)}{x - c} f(x) \right) \\ &= g(c)f'(c) + g'(c)f(c) \end{aligned}$$

We consider (d).

$$\begin{aligned} \lim_{x\to c} \frac{f(x)/g(x) - f(c)/g(c)}{x - c} &= \lim_{x\to c} \frac{f(x)g(c) - f(c)g(x)}{g(x)g(c)(x - c)} \\ &= \lim_{x\to c} \frac{f(x)g(c) - f(c)g(c) + f(c)g(c) - f(c)g(x)}{g(x)g(c)(x - c)} \\ &= \lim_{x\to c} \frac{g(c)(f(x) - f(c))/(x - c) - f(c)(g(x) - g(c))/(x - c)}{g(x)g(c)} \\ &= \frac{g(c)f'(c) - f(c)g'(c)}{g(c)^2} \end{aligned}$$

□

Exercise 2.2. (Leibniz's formula) For a function f and c in the domain of f, define $f^{(2)}(c)$ to be the derivative of f' at c i.e. the derivative of the derivative of f. Similarly, define $f^{(n)}(c)$ to be the derivative of $f^{(n-1)}$ at c. Let $f, g : I \to \mathbb{R}$ be two functions. Suppose $f'(c), f^{(2)}(c), \ldots, f^{(n)}(c)$, and $g'(c), g^{(2)}(c), \ldots, g^{(n)}(c)$ exist.

Show that $(fg)^{(n)}$ is differentiable at c and

$$(fg)^{(n)}(c) = \sum_{i=0}^{n} \binom{n}{i} f^{(n-i)}(c) g^{(i)}(c)$$

2.2. **Composition-The Chain Rule.** We first need a technical result to prove the Chain Rule:

Theorem 2.3. *(Carathéodory's Theorem) Let $f : I \to \mathbb{R}$. Let $c \in I$. Then f is differentiable at c if and only if there exists a function $\phi : I \to \mathbb{R}$ such that ϕ is continuous at c and satisfies*

$$f(x) - f(c) = \phi(x)(x - c).$$

Proof. ($\Rightarrow$) If $f'(c)$ exists, we can define ϕ by:

$$\phi(x) := \begin{cases} \frac{f(x)-f(c)}{x-c} & \text{if } x \neq c \\ f'(c) & \text{if } x = c \end{cases}$$

Since $\lim_{x\to c} \phi(x) = f'(c)$, ϕ is continuous at c.

($\Leftarrow$) If such function ϕ exists, then

$$\frac{f(x) - f(c)}{x - c} = \phi(x)$$

for $x \neq c$. Since ϕ is continuous,

$$\lim_{x\to c} \frac{f(x) - f(c)}{x - c} = \lim_{x\to c} \phi(x)$$

exists. □

Example 2.4. Carathéodory's Theorem can be in more explicit form in some cases:

(1) Let $f(x) = x^3$. Then $\phi(x) = x^2 + cx + c^2$.
(2) Let $f(x) = x^n$. Then $\phi(x) = x^{n-1} + cx^{n-2} + \ldots + c^{n-1}$.

Theorem 2.5. *(Chain Rule) Let I and J be intervals. Let $f : I \to \mathbb{R}$ and let $g : J \to \mathbb{R}$. Suppose $f(I) \subset J$. If f is differentiable at c and g is differentiable at $f(c)$, then $g \circ f$ is differentiable at c and*

$$(g \circ f)'(c) = g'(f(c))f'(c).$$

Proof. By Carathéodory's Theorem, there is a function ψ which is continuous at c such that

$$f(x) - f(c) = \psi(x)(x - c).$$

By Carathéodory's Theorem, there is a function ϕ which is continuous at $f(c)$ such that

$$g(y) - g(f(c)) = \phi(y)(y - f(c)).$$

Thus,

$$g(f(x)) - g(f(c)) = \phi(f(x))\psi(x)(x - c)$$

and so

$$(g \circ f)'(c) = \lim_{x \to c} \frac{g(f(x)) - g(f(c))}{x - c} = \lim_{x \to c} \phi(f(x))\psi(x) = g'(f(c))f'(c).$$

□

Exercise 2.6. In order to understand why one needs Carathéodory's theorem, point out what is the problem of the following proof:

$$\begin{aligned}\lim_{x \to c} \frac{(g \circ f)(x) - (g \circ f)(c)}{x - c} &= \lim_{x \to c} \frac{(g \circ f)(x) - (g \circ f)(c)}{f(x) - f(c)} \cdot \frac{f(x) - f(c)}{x - c} \\ &= \lim_{x \to c} \frac{(g \circ f)(x) - (g \circ f)(c)}{f(x) - f(c)} \cdot \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \\ &= f'(g(c)) \cdot g'(c).\end{aligned}$$

You may consider the following example:

$$f(x) = x^2 \sin(\frac{1}{x}), \quad g(x) = x.$$

2.3. **More examples.**

Exercise 2.7.
(1) Let f and g be two differentiable functions such that the composition $g \circ f(x) = x$. Show that $g'(f(x))f'(x) = 1$.
(2) For $m \in \mathbb{N}$, let $f(x) = x^{\frac{1}{m}}$. Use above to deduce that $f'(x) = \frac{1}{m}x^{\frac{1}{m}-1}$.
(3) For $n, m \in \mathbb{N}$, let $h(x) = x^{\frac{n}{m}}$. Write down a formula for $h'(x)$.

Exercise 2.8.
(a) We showed before that

$$f(x) = \begin{cases} x \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

is continuous. From previous discussions, we know f is differentiable at $c \neq 0$. Is f differentiable at 0?

(b) How about if one considers

$$g(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

What can you say about the continuity of g'?

3. Relative Extremum

Definition 3.1. (1) A function $f : I \to \mathbb{R}$ is said to have a relative maximum at c if there exists $\delta > 0$ such that $f(x) \leq f(c)$ for all $x \in I \cap (c - \delta, c + \delta)$.
(2) A function $f : I \to \mathbb{R}$ is said to have a relative minimum at c if there exists $\delta > 0$ such that $f(x) \geq f(c)$ for all $x \in I \cap (c - \delta, c + \delta)$.
(3) We say that a function f has a relative extremum if f has either a relative maximum or a relative minimum.

Exercise 3.2. Show that an absolute maximum of a function f is a relative maximum.

Example 3.3. (1) Let $f : [0, 6] \to \mathbb{R}$ given by $f(x) = -x^2 + 4x + 5$. Find all relative maximum point(s) of f and all relative minimum point(s) of f (if exists).
(2) Let $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = \sin(x)$. Find all relative maximum point(s) of f and all relative minimum point(s) of f.

Exercise 3.4. Let $f : [0 \to \infty) \to \mathbb{R}$ given by

$$f(x) = x \sin(x).$$

Prove that f has infinitely many relative maxima, but does not have an absolute maximum.

A point c is said to be an interior point in a set $A \subset \mathbb{R}$, if there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subset A$. In particular, an interior point must be a point in A. For example, in the interval $[0, 1]$, the points in $(0, 1)$ are the interior points of A, and $0, 1$ are not interior points of A.

Theorem 3.5. *Let c be an interior point in A at which f has a relative extremum. If $f : A \to \mathbb{R}$ is differentiable at c, then $f'(c) = 0$.*

Proof. We consider the case that f has a relative maximum (and the case that f has a relative minimum is similar). Suppose $f'(c) > 0$. Then there exists $\delta > 0$ such that for $x \in (c - \delta, c + \delta)$,

$$\frac{f(x) - f(c)}{x - c} > 0.$$

Then, for $x \in (c - \delta, c + \delta)$ and $x > c$, $f(x) > f(c)$, giving a contradiction. Now, one uses a similar argument to show $f'(c) < 0$ is impossible. Hence, we must have that $f'(c) = 0$. □

Example 3.6. Let $f : [0, 5] \to \mathbb{R}$, $f(x) = x^2 - 8x + 5$. Then f has a relative minimum at the interior point 4, and $f'(4) = 2(4) - 8 = 0$. Note that f also has an absolute (and so relative) maximum at 0, but 0 is not an interior point.

Exercise 3.7. Could you explain how one can locate the relative maximum in Exercise 3.4? Show that $f(x) = x\sin(x)$ has the relative maxima at some points in $(2k\pi + \frac{\pi}{2}, 2k\pi + \pi)$ for $k \geq 0$, and has the relative minima at some points in $(2k\pi + \frac{3\pi}{2}, 2k\pi + 2\pi)$.

Example 3.8. As you know from Calculus, the converse of Theorem 3.5 is false, and one needs additional conditions (so-called First-Derivative Test and Second Derivative Test discussed later) to determine a relative extremum.

The typical example is $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^3$. In such case, f is increasing and so does not have a relative extremum. But, $f'(0) = 2(0)^2 = 0$.

Exercise 3.9.
(1) Rephrase Theorem 3.5 as follows: Let $f : I \to \mathbb{R}$. If has ether a relative maximum or a relative minimum at c, then either f is not differentiable at c or $f'(c) = 0$ or c is not an interior point.
(2) Give an example of a function which has a relative maximum at 0, and is not differentiable at 0.

4. The Mean Value Theorem

Remark 4.1. In the following theorems, we usually consider a function $f : [a, b] \to \mathbb{R}$ such that f is continuous on $[a, b]$ and differentiable on (a, b). One may ask if there is an example that is continuous on $[a, b]$ and differentiable on (a, b), but not differentiable at a or b. In fact, one such example is given in Exercise 2.8(a).

Theorem 4.2. *(Rolle's Theorem) Let $f : [a, b] \to \mathbb{R}$. Suppose f is continuous on $[a, b]$, and is differentiable at all points on (a, b). If $f(a) = f(b) = 0$, then there exists a point $c \in (a, b)$ such that $f'(c) = 0$.*

Proof. If f is a constant function, then $f'(c) = 0$ for all $c \in (a, b)$. Then we are done.

We now consider f is not constant. Suppose f attains some positive values in (a, b). By the Maximum-Minimum Theorem, f has an absolute maximum on $[a, b]$. Since $f(a) = f(b) = 0$, f attains the positive absolute maximum at some $c \in (a, b)$. By Theorem 3.5 and the fact that c is an interior point in $[a, b]$, $f'(c) = 0$.

Suppose f does not attain some positive values in (a, b). Then $-f$ attains some positive values in (a, b). Then above discussion gives that $(-f)'(c) = 0$. Hence, $f'(c) = 0$. □

Theorem 4.3. *(Mean Value Theorem) Let $f : [a, b] \to \mathbb{R}$. Suppose f is continuous on $[a, b]$ and is differentiable on (a, b). Then there exists at least one $c \in (a, b)$ such that*

$$f(b) - f(a) = f'(c)(b - a).$$

Proof. Define

$$\psi(x) := f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Note that $\psi(a) = \psi(b) = 0$. Moreover, ψ is continuous on $[a, b]$ and is differentiable on (a, b). Then, by Rolle's Theorem, there exists $c \in (a, b)$ such that $\psi'(c) = 0$. Hence,

$$0 = \psi'(c) = f'(c) - \frac{f(b) - f(b)}{b - a}$$

and so $f(b) - f(a) = f'(c)(b - a)$. □

Exercise 4.4. Draw a curve $y = f(x)$ in the interval $[a, b]$. Interpret $\frac{f(b)-f(a)}{b-a}$ as a slope between points $(a, f(a))$ and $(b, f(b))$, interpret $f'(c)$ as the slope of the tangent to $y = f(x)$ at a point $(c, f(c))$.

Exercise 4.5. Do you see why Rolle's Theorem (Theorem 4.2) is a special case of the Mean Value Theorem (Theorem 4.3)?

Exercise 4.6. (Applications on inequalities)

(1) Use the Mean Value Theorem to show that, for any $x, y \in \mathbb{R}$,

$$|\sin(x) - \sin(y)| \leq |x - y|$$

(2) Use the Mean Value Theorem to show that for $x \in (0, \frac{\pi}{2})$,

$$1 < \frac{\tan(x)}{x}.$$

We can use the Mean-Value Theorem to show that a function with zero derivative must be a constant function:

Theorem 4.7. *(zero derivative ⇒ constant function) Suppose $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and is differentiable on (a, b). If $f'(x) = 0$ for all $x \in (a, b)$, then $f(x)$ is a constant function on $[a, b]$.*

Proof. Let $x \in [a, b]$. We apply the Mean-Value Theorem on the interval $[a, x]$, and so

$$f(x) - f(a) = f'(c)(x - a)$$

for some $c \in (a, x)$. Since $f'(c) = 0$, $f(x) = f(a)$ □

Corollary 4.8. *Let $f, g : [a, b] \to \mathbb{R}$. Suppose f and g are continuous on $[a, b]$ and differentiable on (a, b). If $f'(c) = g'(c)$ for all $c \in (a, b)$, then $f(x) = g(x) + C$ for some constant C.*

5. More applications of the MVT on monotone functions and extrema

5.1. **Increasing and decreasing functions.** A function $f : I \to \mathbb{R}$ is said to be increasing if for any $x_1, x_2 \in I$ satisfying $x_1 < x_2$, $f(x_1) \leq f(x_2)$. A function $f : I \to \mathbb{R}$ is said to be decreasing if $-f$ is increasing.

Theorem 5.1. *Suppose $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and is differentiable function on (a, b). Then*

(a) *f is increasing on $[a, b]$ if and only if $f'(x) \geq 0$ for all $x \in (a, b)$;*
(b) *f is decreasing on $[a, b]$ if and only if $f'(x) \leq 0$ for all $x \in (a, b)$.*

Proof. We consider (a).
If direction (by using Mean-Value Theorem): Suppose $f'(x) \geq 0$ for all $x \in I$. Let $x_1 < x_2$ in $[a, b)$. By the Mean-Value Theorem,

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

for some $c \in (x_1, x_2)$. Since $f'(c) \geq 0$ and $x_2 - x_1 > 0$, we have $f(x_2) \geq f(x_1)$. Since x_1 and x_2 are arbitrary, f is increasing on $[a, b)$. It is also similar too show f is also increasing for $(a, b]$. Then f is increasing on $[a, b]$.

Only if direction (by taking limit): Suppose f is an increasing function. Let $c \in (a, b)$. Then, for any $x \neq c$,

$$\frac{f(x) - f(c)}{x - c} \geq 0$$

Hence, $\lim_{x \to c} \frac{f(x)-f(c)}{x-c} \geq 0$ and so $f'(c) \geq 0$.

For (b), if follows from the following equivalent statements:

(i) f is decreasing;
(ii) $-f$ is increasing;
(iii) $(-f)'(x) \geq 0$ for all $x \in I$;
(iv) $f'(x) \leq 0$ for all $x \in I$.

(i)⇔(ii) follows from definitions. (ii)⇔ follows from (a). (iii)⇔(iv) follows from $(-f)'(x) = -f'(x)$. □

Remark 5.2. Let I be an interval. Let $f : I \to \mathbb{R}$. Let $c \in I$. One may hope for that: If $f'(c) > 0$, then there exists $\delta > 0$ such that $f(x)$ is increasing on $(c-\delta, c+\delta)$. Unfortunately, it is false. Indeed, if one wants to have a counterexample, one must find a function whose derivative is not continuous at c (Why?). Here is one:

$$f(x) = \begin{cases} x + 2x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Example 5.3. (Applications on proving inequalities) Show, by using Theorem 5.1 that

(a) for any $x \geq 0$, $\sin(x) \leq x$.
(b) for any $x \in \mathbb{R}$, $ex \leq e^x$.

5.2. **First derivative test.**

Theorem 5.4. *(First derivative test) Let $f : (a, b) \to \mathbb{R}$ be a continuous function. Let $c \in (a, b)$.*

(a) *If there exists $\delta > 0$ such that $f'(x) \geq 0$ for $x \in (c - \delta, c)$ and $f'(x) \leq 0$ for $x \in (c, c + \delta)$, then f has a relative maximum at c.*
(b) *If there exists $\delta > 0$ such that $f'(x) \leq 0$ for $x \in (c - \delta, c)$ and $f'(x) \geq 0$ for $x \in (c, c + \delta)$, then f has a relative minimum at c.*

Proof. We only prove for (a), and a proof for (b) is similar. It follows from Theorem 5.1 that f is increasing from $(c-\delta, c]$ and decreasing on $[c, c+\delta)$. Thus, $f(c) \geq f(x)$ for all $x \in (c - \delta, c + \delta)$. This implies that f has a relative maximum at c.

□

Example 5.5. Let $f : (-\infty, 0) \cup (0, \infty) \to \mathbb{R}$ given by $f(x) = x + \frac{1}{x}$. Find the point(s) of relative maximum and relative minimum of f.

Example 5.6. Give an example that the first derivative test in Theorem 5.4 can be applied to a situation that $f'(c)$ does not exist.

6. Darboux's Theorem

In order to motivate Darboux's theorem, one may try to show the following simpler version of the theorem:

Exercise 6.1. Let $f : [a, b] \to \mathbb{R}$. Suppose f is differentiable on $[a, b]$ and the derivative of f is continuous on $[a, b]$. Let k be a number between $f'(a)$ and $f'(b)$. Then there is $c \in [a, b]$ such that $f'(c) = k$.

However, we have seen in Exercise 2.8 that the derivative may not be continuous. We shall prove the intermediate value theorem for derivatives in Theorem 6.3.

Lemma 6.2. *Let $f : [a, b] \to \mathbb{R}$. Let $c \in [a, b]$. Suppose the derivative of f exists at c. Then*

(a) *If $f'(c) > 0$, then there exists $\delta > 0$ such that $f(x) > f(c)$ for $x \in [a, b]$ satisfying $c < x < c + \delta$.*
(b) *If $f'(c) < 0$, then there exists $\delta < 0$ such that $f(x) > f(c)$ for $x \in [a, b]$ satisfying $c - \delta < x < c$.*

Proof. Exercise. See the proof of Theorem 5.1. □

Theorem 6.3. *(Darboux's Theorem/ Intermediate Value Theorem for Derivatives) Let $f : [a, b] \to \mathbb{R}$ be differentiable. Let k be a number between $f'(a)$ and $f'(b)$. Then there is at least one $c \in (a, b)$ such that $f'(c) = k$.*

Proof. (Use relative extremum at $c \Rightarrow$ zero derivative at that point)

We shall only prove the case that $f'(a) < k < f'(b)$ (the case that $f'(b) < k < f'(a)$ is similar). Let $g(x) = kx - f(x)$. Then $g'(x) = k - f'(x)$. Hence, $g'(a) > 0$ and $g'(b) < 0$. By Lemma 6.2 (a) and (b), g does not attain the absolute maximum at a and b respectively.

The Maximum-Minimum Theorem guarantees that g attains some absolute maximum and so g attains the absolute maximum at some $c \in (a, b)$. Now, by Theorem 3.5, $g'(c) = 0$. Thus, $f'(c) = k$. □

Example 6.4. Recall that

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

f' is not continuous at 0 (but exists at 0). Thus f' gives an example a non-continuous function that the Bolzano's intermediate value theorem (for continuous functions) still works.

Example 6.5. Let

$$g(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Then, Darboux's theorem says that there is no differentiable function $f : \mathbb{R} \to \mathbb{R}$ such that $f'(x) = g(x)$.

7. Taylor's Theorem and Taylor's Series

7.1. **Motivation.** Let $f : (a, b) \to \mathbb{R}$. Suppose f is continuous and differentiable on (a, b). Then, one obtains a function of the derivative $f' : (a, b) \to \mathbb{R}$. Then, we can consider the derivative of f' at $c \in (a, b)$. If such derivative at c exists, then denote such number by $f''(c)$ or $f^{(2)}(c)$, called the second derivative of f at c. In similar fashion, we obtain the third derivative at c, denoted by $f^{(3)}(c)$, and in general, the n-th derivative at c, denoted by $f^{(n)}(c)$. As convention, $f^{(0)} = f$.

Exercise 7.1. Let $f : (a, b) \to \mathbb{R}$. Let $c \in (a, b)$. Suppose $f'(c), \ldots, f^{(n)}(c)$ exist.

(1) Let

$$P_n(x) = f(c) + f'(c)(x - c) + \frac{f^{(2)}(c)}{2!}(x - c)^2 + \ldots + \frac{f^{(n)}(c)}{n!}(x - c)^n.$$

Show that, for all $k = 0, \ldots, n$, $P_n^{(k)}(c) = f^{(k)}(c)$.

(2) There exists a unique polynomial $P(x)$ of degree n such that for $k = 0, 1, \ldots, n$, $P^{(k)}(c) = f^{(k)}(c)$.

The goal of Taylor's theorem is to approximate a function $f(x)$ by some polynomials around a point c, and polynomials are much more computable by hand and so are useful for numerical analysis.

7.2. **Taylor's Theorem.**

Theorem 7.2. *Let $f : (a, b) \to \mathbb{R}$. Suppose f and the derivatives $f^{(1)}, f^{(2)}, \ldots, f^{(n-1)}, f^{(n)}$ exist and are continuous on (a, b). Fix $x_0 \in (a, b)$. Then, for any $x \in (a, b)$, there exists c in between x and x_0 such that*

$$f(x) = f(x_0)+f'(x_0)(x-x_0)+\frac{f^{(2)}(x_0)}{2!}(x-x_0)^2+\ldots+\frac{f^{(n-1)}(x_0)}{(n-1)!}(x-x_0)^n+\frac{f^{(n)}(c)}{n!}(x-x_0)^n.$$

Exercise 7.3. We shall only ontline how to work out the proof of Theorem 7.2 for $n = 1, 2$.

(a) When $n = 1$, show that the statement follows from the Mean Value Theorem.

(b) Consider $n = 2$. Let

$$g(t) = f(x_0) + f'(x_0)(t - x_0) + \frac{M}{2!}(t - x_0)^2 - f(t),$$

where $M \in \mathbb{R}$ is chosen such that $g(x) = 0$.

(i) Find $g'(t)$ and $g^{(2)}(t)$.

(ii) Show $g(x) = g(x_0) = 0$, and $g'(x_0) = 0$.

(iii) Use (i) and (ii) to show that $M = g^{(2)}(c)$ for some c in between x and x_0.

Definition 7.4. We use the notations in Theorem 7.2. The n-th Taylor's polynomial (for f about x_0) is

$$T_n(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k.$$

The n-th remainder term (for f about x_0) is defined as:

$$R_n(x) = f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

7.3. **Limit of the remainder $R_n(x)$ and Taylor's series.**

Theorem 7.5. *Let $f : (a, b) \to \mathbb{R}$. Fix $x_0 \in (a, b)$. Suppose derivatives $f^{(k)}(x)$ exist for all k and for all $x \in (a, b)$. Moreover, suppose all $f^{(k)}$ are bounded by a*

single constant M i.e. for all integers $k \geq 0$ and $x \in (a, b)$,

$$|f^{(k)}(x)| \leq M$$

Then,

$$\lim_{n\to\infty} R_n(x) = 0$$

for all $x \in (a, b)$.

Proof. Let $x \in (a, b)$. By Taylor's Theorem,

$$R_n(x) = \frac{f^{(n)}(c)}{n!}(x - x_0)^n$$

for some c between x and x_0. Hence,

$$|R_n(x)| = \frac{|f^{(n)}(c)|}{n!}|(x - x_0)^n| \leq \frac{M}{n!}|x - x_0|^n$$

Since

$$\lim_{n\to\infty} \frac{M}{n!}|x - x_0|^n = 0$$

the squeeze theorem gives that

$$\lim_{n\to\infty} |R_n(x)| = 0.$$

□

Corollary 7.6. *Let f be as in Theorem 7.5. Then, for all $x \in (a, b)$,*

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

Proof. By taking the limit $n \to \infty$ on

$$R_n(x) = f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k,$$

we have the corollary. □

Definition 7.7. We shall call

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

to be the Taylor series (or Taylor Series expansion) of f about x_0.

7.4. **Examples.**

Example 7.8. (Exponential functions) Let $f(x) = e^x$. Then $f^{(k)}(x) = e^x$ for all $x \in \mathbb{R}$ and all k. We fix some $a < 0$ and some $b > 0$, and consider f in the interval $[a, b]$. For all $x \in (a, b)$, and for all $k \geq 0$,

$$|f^{(k)}(x)| < e^b.$$

Thus,

$$\begin{aligned} e^x =& 1 + \frac{f^{(1)}(0)}{1!}x + \frac{f^{(2)}(0)}{2!}x^2 + \dots \\ =& 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots \\ =& \sum_{k=0}^{\infty} \frac{x^k}{k!} \end{aligned}$$

Example 7.9. Let $f(x) = \cos(x)$. Then

$$f^{(k)}(x) = \begin{cases} -\sin(x) & k = 1, 5, 9, 13, \dots \\ -\cos(x) & k = 2, 6, 10, 14, \dots \\ \sin(x) & k = 3, 7, 11, 15, \dots \\ \cos(x) & k = 4, 8, 12, \dots \end{cases}$$

All above functions are bounded on $\mathbb{R}$. Thus,

$$\cos(x) = \sum_{k=0}^{n} \frac{(-1)^k}{(2k)!}x^{2k}. \tag{7.1}$$

Exercise 7.10. Let $f(x) = \ln(1 + x)$.

(a) Find $f^{(k)}(x)$.
(b) Find the n-th Taylor polynomial $T_n(x)$ for f about 0.
(c) Find the remainder term $R_n(x)$ for f about 0.
(d) Find $\lim_{n\to\infty} R_n(1)$. (Note that you cannot apply Theorem 7.5 in this example)
(e) Use above and Taylor's theorem to show that

$$\ln(2) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$$

Exercise 7.11. Let $f : \mathbb{R} \to \mathbb{R}$ given by:

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}.$$

Show that the Taylor series expansion of f about 0 is identically 0, and so is not equal to f.